CSN301 | SOFTWARE ENGINEERING | April 17, 2026
Design & Implementation of Online Examination System ||

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Project Chosen: Design & Implementation of Online Examination System
Project Number : **P17**

For this project, you will solve problems based on what you have learned in **this course.**

Instructions ***// These are sample instructions, you may keep relevant ones and remove/add rest as required.***

- The project must be developed using appropriate platform for Design and Code.
- Mention names and IDs of all group members on the top of this document.
- There are ___ questions in this assignment. Attempt any one of them, and mention which project you have chosen on the top of this page. In each question, objective to be achieved, expected inputs and expected outputs are mentioned.
- **Project submitted after due date will not be evaluated and a score of zero will be awarded.**
- Upload a **word version** of the document.
- The group size should not be more than 3, and the group must be formed from the same section. Inter section groups will not be accepted.
- Plagiarism may lead to award of zero marks.
- If you have doubts about any project, please contact the course instructor.

Due Date: **12 pm, April 17, 2026.**

Submitting this Assignment

You will submit (upload) this assignment in LMS. Email/paper submissions will not be accepted.

- Write your answer (codes and snapshots of module interface, input and output) after the question in this document.
- Name this document as Project_SEEVEN2026_SAP1_SAP2_SAP3.doc in case group members IDs are SAP1, SAP2 and SAP3 respectively, acronym of course is SE, Sem is EVEN, and year is 2026.

Grading Scheme: This project has 10 marks. Marks distribution within this project is as follows //(Faculty may change this text and distribution as deemed fit):

- Appropriate GUI for Module interface, input and output as mentioned in the question: X marks,
- Expected input and Output as mentioned in the question: Y marks,
- Well documented code, Presentation = Z marks.

// Where X+Y+Z = 10

ACKNOWLEDGMENT

We would like to express our sincere gratitude to **Dr. Dhruva Chaudhary**, Assistant Professor, School of Computing, DIT University, for his valuable guidance and support throughout this project. His insightful feedback and encouragement have been instrumental in the successful completion of our work.

We also extend our appreciation to DIT University for providing us with the necessary resources and a conducive learning environment to work on this project.

Lastly, we would like to acknowledge the efforts and dedication of our team members,

VANSH VERMA , AKANKSHA , SUMIT KR SINGH whose collaborative efforts made this project possible.

Project Title :-

P17 : Design & Implementation of Online Examination System

Submitted By:

Name: Akanksha	SAP ID: 1000021006	Section: P
Name: Vansh Verma	SAP ID: 1000020982	Section: P
Name: Sumit kr Singh	SAP ID: 1000021037	Section: P

Submitted To:

Dr. Dhruva Chaudhary
Assistant Professor
School of Computing, DIT University

CANDIDATE DECLARATION

We, the undersigned, hereby declare that the project titled **"P17: Design & Implementation of Online Examination System"** is our original work. This project has been carried out under the guidance of **Dr. Dhruva Chaudhary**, Assistant Professor, School of Computing, DIT University.

We affirm that this project has not been submitted to any other university or institution for the award of any degree or diploma. All references and sources used in this work have been duly acknowledged.

We take full responsibility for the authenticity, accuracy, and integrity of the content presented in this report. We understand that any violation of the above statements may lead to disciplinary action as per university rules.

Submitted By:
Vansh Verma
Akanksha
Sumit Kr Singh

Submitted To:
Dr. Dhruva Chaudhary
Assistant Professor
School of Computing, DIT University

Project Description

Project Title: P17 – Design & Implementation of Online Examination System

Objective:

The main objective of this project is to design and develop a secure, efficient, and user-friendly Online Examination System using Java. The system aims to automate the examination process by providing a digital platform for conducting exams, managing questions, evaluating answers, and generating results in real time.
It focuses on reducing manual effort, ensuring accuracy in evaluation, and providing a seamless experience for both students and administrators.

Problem Statement:

Traditional examination systems are time-consuming, resource-intensive, and prone to human errors in evaluation and result processing. Managing large numbers of students, maintaining records, and ensuring fair assessment becomes difficult in manual systems.

Additionally, there is a need for a secure and scalable platform that allows remote access, prevents malpractice, and ensures data integrity during examinations.

The Online Examination System addresses these challenges by providing a centralized, automated solution that enables efficient exam management, secure access, and instant result generation.

Project Overview:

The Online Examination System is a Java-based application designed to conduct exams in a digital environment. It provides separate modules for administrators and students, ensuring proper role-based access and control.

The system enables users to:

• Register and login securely using authentication mechanisms.
• Allow administrators to create, update, and manage question banks.
• Conduct timed examinations with automatic submission.
• Display multiple-choice questions (MCQs) in an organized format.

• Evaluate answers automatically and calculate scores instantly.
• Store student responses and results in a structured database.
• Provide performance feedback and result analysis.
• Prevent unauthorized access using basic validation and session control.

Source Code:

The following Java classes/modules are part of the project:

• User.java – Handles user details such as login credentials and roles (Admin/Student).

• Login.java – Manages authentication and validation of users.
• Question.java – Represents question structure including options and correct answers.
• Exam.java – Controls exam flow, timing, and question navigation.
• Result.java – Calculates and stores exam results and scores.
• AdminPanel.java – Provides functionalities for managing questions and exams.
• StudentPanel.java – Allows students to take exams and view results.
• DatabaseManager.java – Handles database connectivity and operations.
• MainFrame.java – Main GUI window for navigation.
• Main.java – Entry point of the application.

Technology Stack:

• Core Technology: Java
• User Interface (GUI): Java Swing / AWT
• Event Handling: Java Event Handling Model
• Database: MySQL / SQLite (via JDBC)
• Authentication: Username & Password-based login system
• Data Storage: Relational Database Management System
• Development Environment: VS Code / IntelliJ IDEA
• Java Version: JDK 17 / JDK 22

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed description of the **Online Examination System**. It outlines the system's functionality, requirements, constraints, and overall behavior.
This document is intended for developers, project evaluators, and stakeholders to understand how the system operates and what features it provides.

1.2 Scope

The Online Examination System is designed to automate the process of conducting examinations in an educational institution. It allows students to register, log in, take exams online, and receive instant results.

The system also enables administrators to manage question banks, schedule exams, evaluate responses automatically, and maintain records efficiently.

This system reduces manual effort, increases accuracy, ensures quick evaluation, and provides a secure platform for conducting exams.

1.3 Definitions

- **User**: A person interacting with the system (Student/Admin)
- **Exam**: A test consisting of multiple questions
- **Question Bank**: Collection of questions stored in the system
- **Result**: Score obtained by a student after evaluation

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2. Functional Requirements

2.1 Student Registration

- The system shall allow new users to register by entering details such as name, email, and password.

- The system shall validate input fields before registration.
- The system shall store user data securely in the database.

2.2 User Login

- The system shall allow registered users to log in using valid credentials.
- The system shall verify username and password from the database.
- The system shall restrict access for invalid login attempts.

2.3 Exam Management

- The system shall allow administrators to create exams.
- The system shall allow setting exam duration and total marks.
- The system shall allow scheduling exams.

2.4 Question Bank Management

- The system shall allow administrators to add, edit, and delete questions.
- Each question shall have multiple options and one correct answer.
- Questions shall be stored in a structured database.

2.5 Taking Examination

- The system shall display questions to students.
- The system shall allow students to select answers.
- The system shall provide a timer for the exam.
- The system shall auto-submit when time expires.

2.6 Submission of Exam

- The system shall allow students to submit answers manually.
- The system shall save responses in the database.

2.7 Automatic Evaluation

- The system shall compare student answers with correct answers.
- The system shall calculate scores automatically.
- The system shall generate results instantly.

2.8 Result Management

- The system shall display results to students.
- The system shall allow admin to view all results.
- The system shall store results for future reference.

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3. Non-Functional Requirements

3.1 Security

- The system shall ensure secure login using authentication.
- The system shall protect user data from unauthorized access.
- The system shall prevent cheating using session control.

3.2 Performance

- The system shall respond quickly to user actions.
- The system shall handle multiple users simultaneously.
- The system shall ensure fast result generation.

3.3 Usability

- The system shall have a user-friendly interface.

- The system shall be easy to navigate for students and admins.
- The system shall provide clear instructions during exams.

3.4 Reliability

- The system shall work without failure during exams.
- The system shall ensure data is not lost during submission.

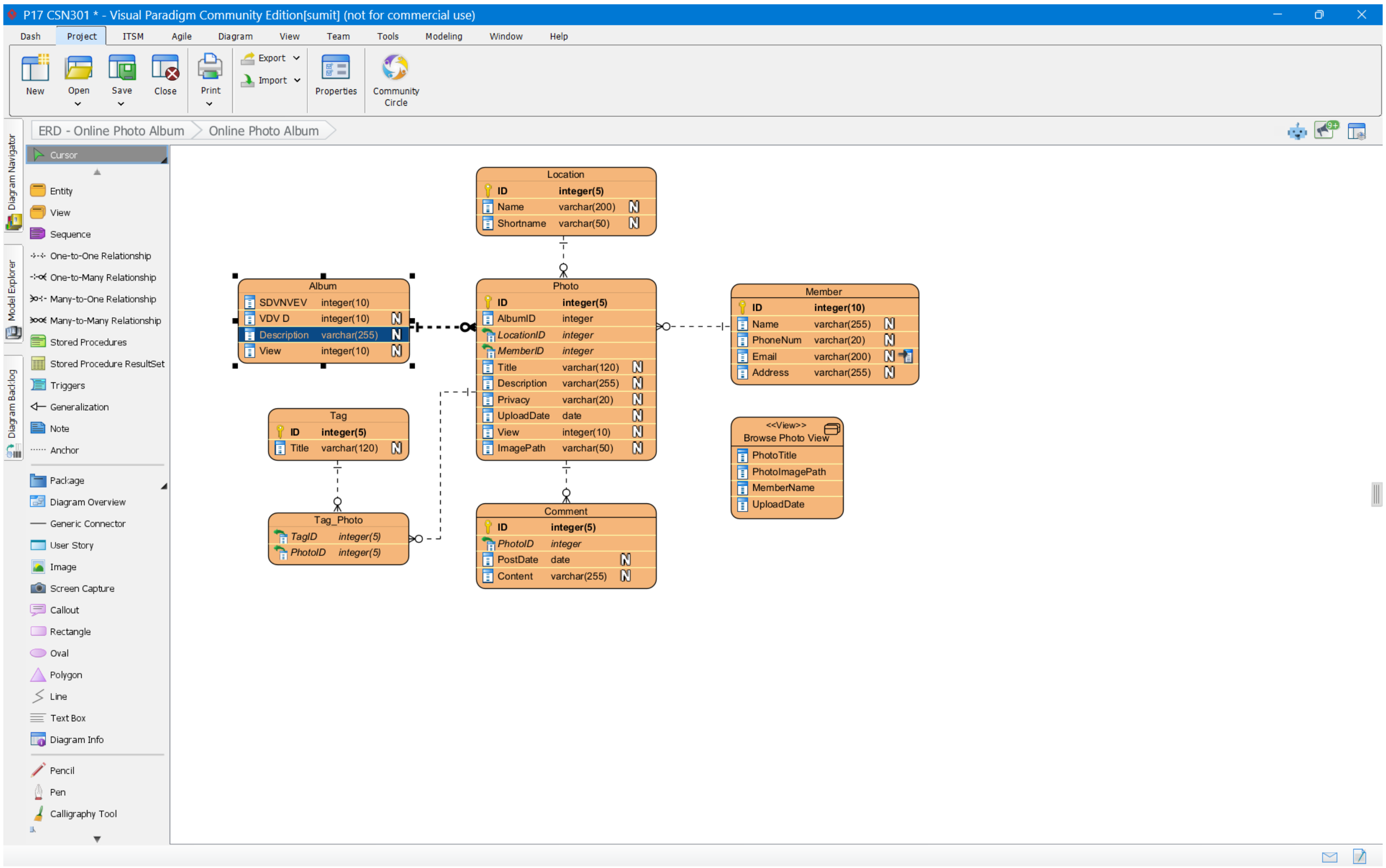
3.5 Maintainability

- The system shall allow easy updates and modifications.
- The system shall support bug fixing and enhancements.

3.6 Availability

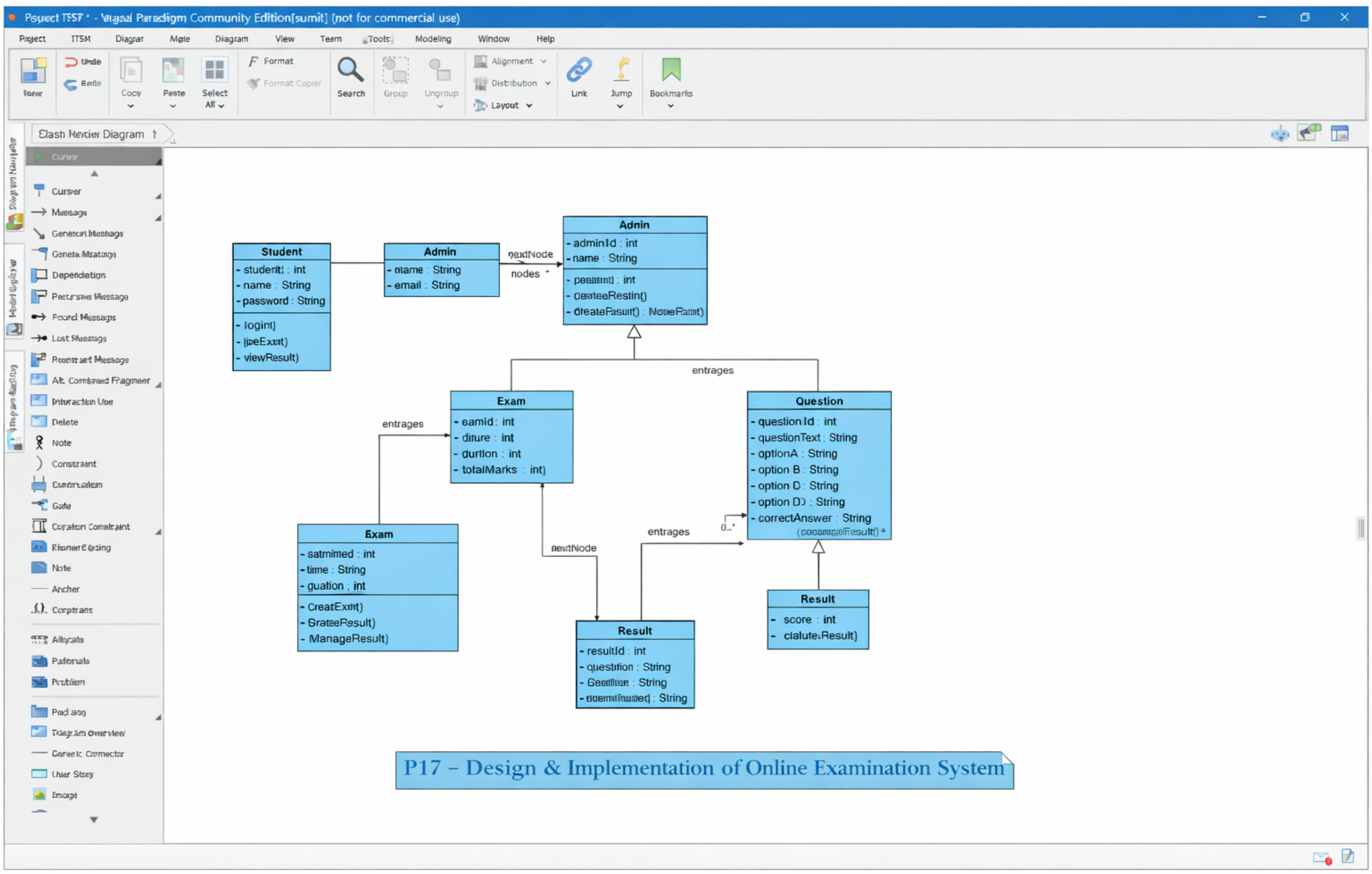
- The system shall be available whenever required by users.
- The system shall minimize downtime.

ENTITY RELATIONSHIP DIAGRAM



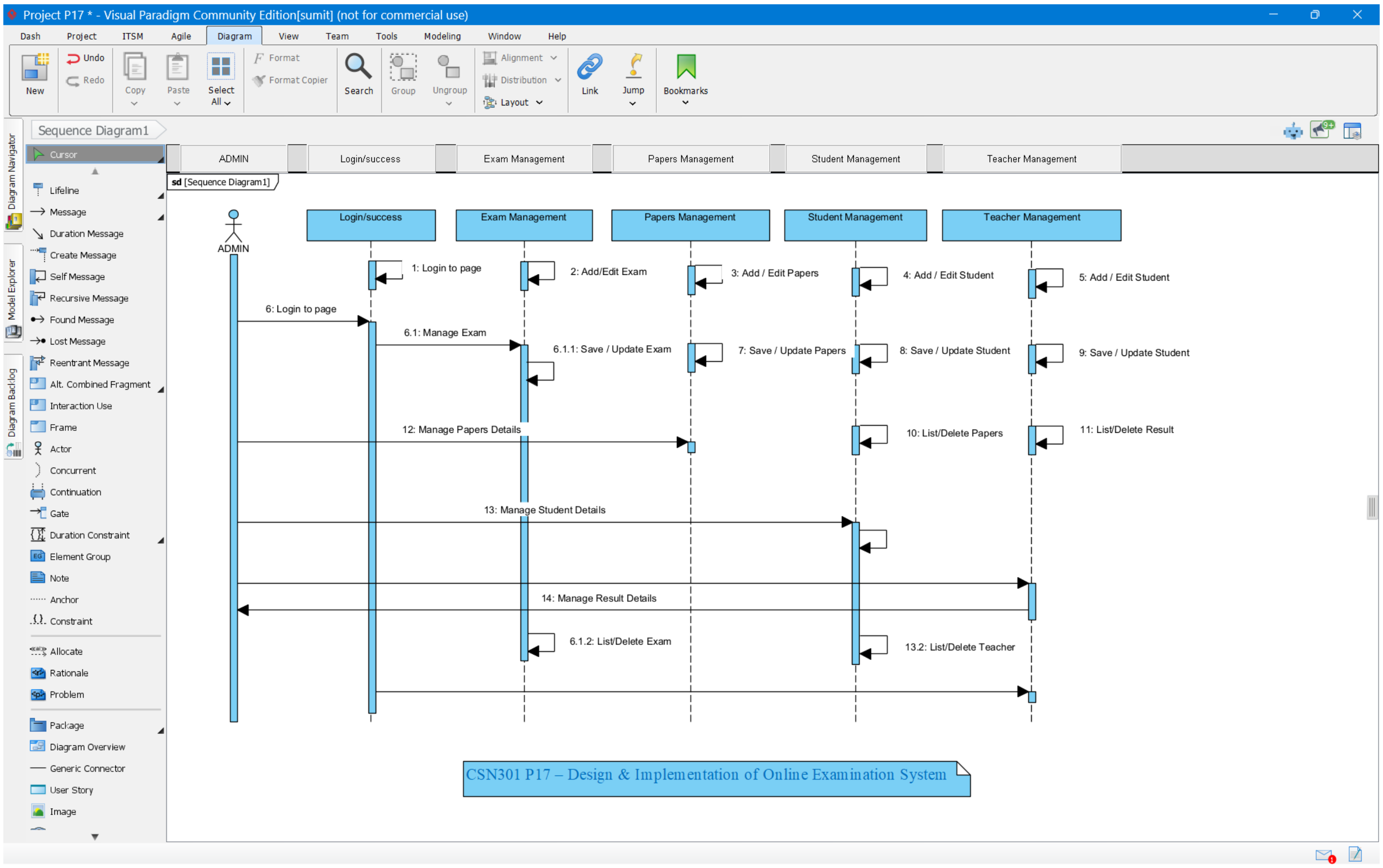
The ER Diagram is used to represent the database structure of the Online Examination System. It shows the entities like Student, Exam, Question, and Result along with their attributes and relationships. This helps in designing and understanding how data is stored and connected in the system.

CLASS DIAGRAM



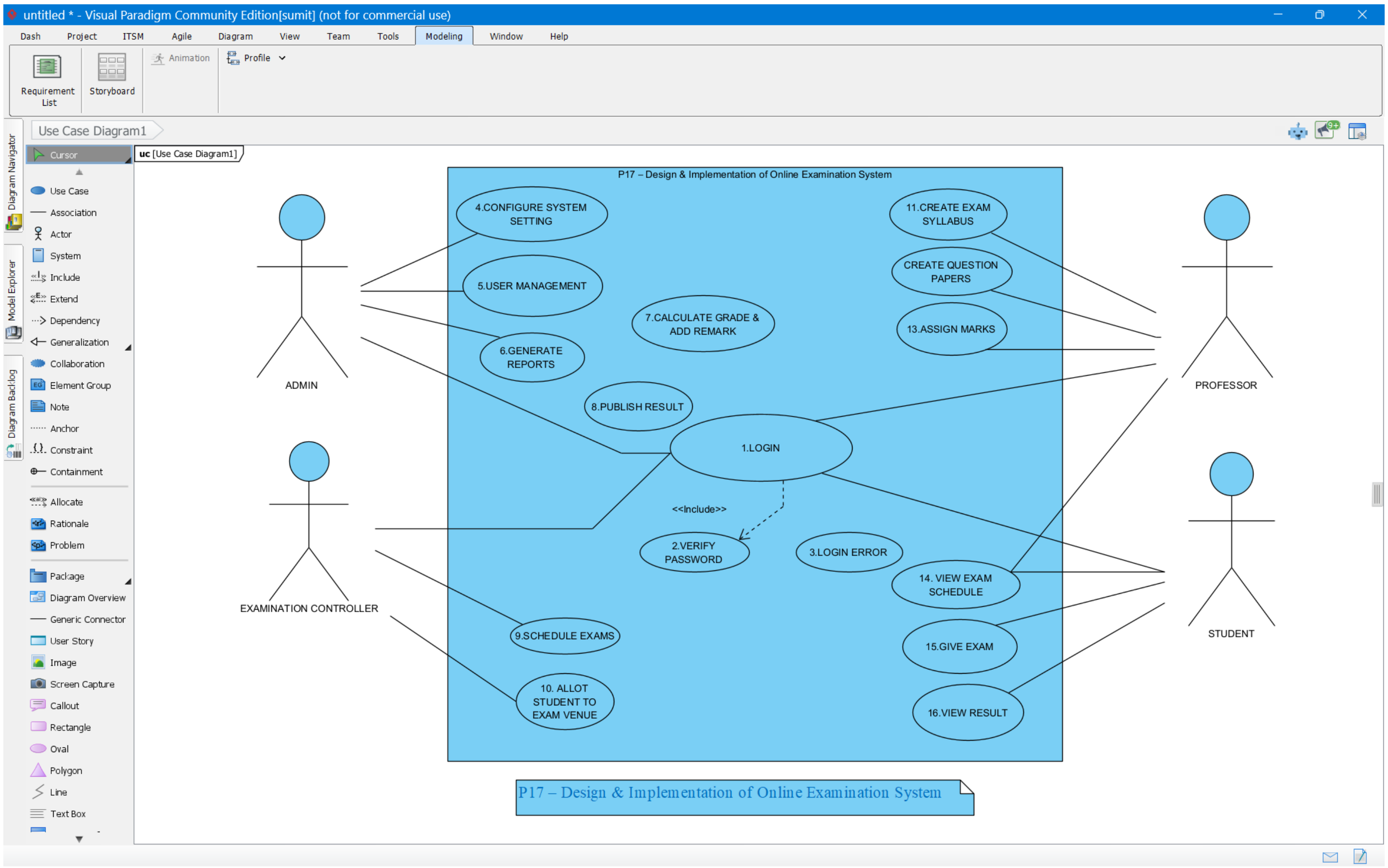
The Class Diagram represents the structure of the Online Examination System by showing classes, their attributes, methods, and relationships. It helps in understanding how different components of the system are organized and interact with each other.

SEQUENCE DIAGRAM



The Sequence Diagram represents the step-by-step flow of interactions between different components of the Online Examination System over time. It shows how the student logs in, takes the exam, submits answers, and how the system processes and returns the result. This diagram helps in understanding the dynamic behavior of the system.

USE CASE DIAGRAM



The Use Case Diagram represents the interaction between users (Student and Admin) and the Online Examination System. It shows the different functionalities like login, taking exams, managing questions, and viewing results.

SYSTEM IMPLEMENTATION (ADD THIS SECTION)

6.1 Technologies Used

- Java
- Swing
- MySQL / JDBC

6.2 Source Code (IMPORTANT)

Add:

- Main.java

```java
public class Main {
    public static void main(String[] args) {
        new Login(); // Opens login screen
    }
}
```

- Login.java

```java
if(username.equals("admin") && password.equals("1234")) {
    JOptionPane.showMessageDialog(null, "Login Successful");
    new Exam(); // Open exam screen
} else {
    JOptionPane.showMessageDialog(null, "Invalid Credentials");
}
```

- Exam.java

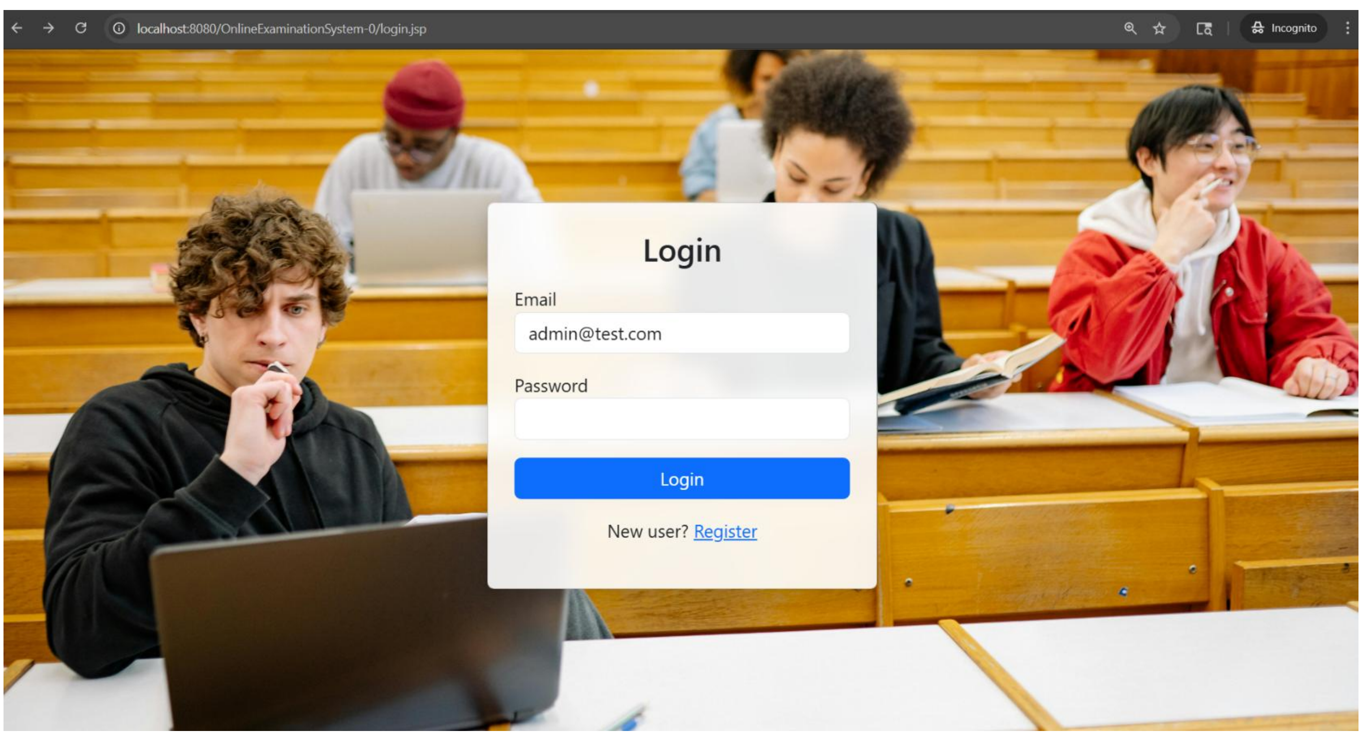
```java
if(selectedAnswer.equals(correctAnswer)) {
    score++;
}

if(timeOver) {
    submitExam();
}
```

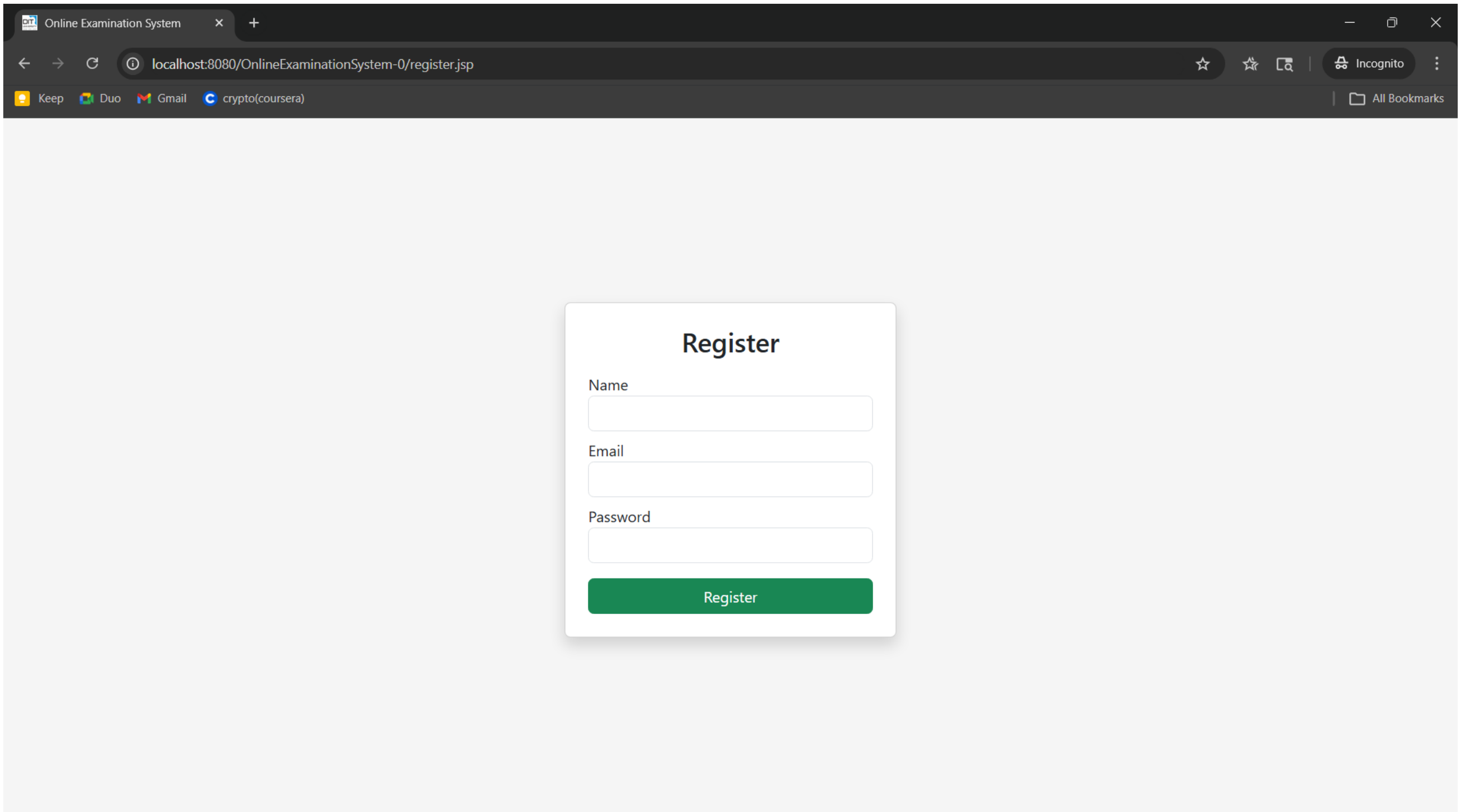
6.3 Output Screens / UI

Add screenshots of:

- Login Page



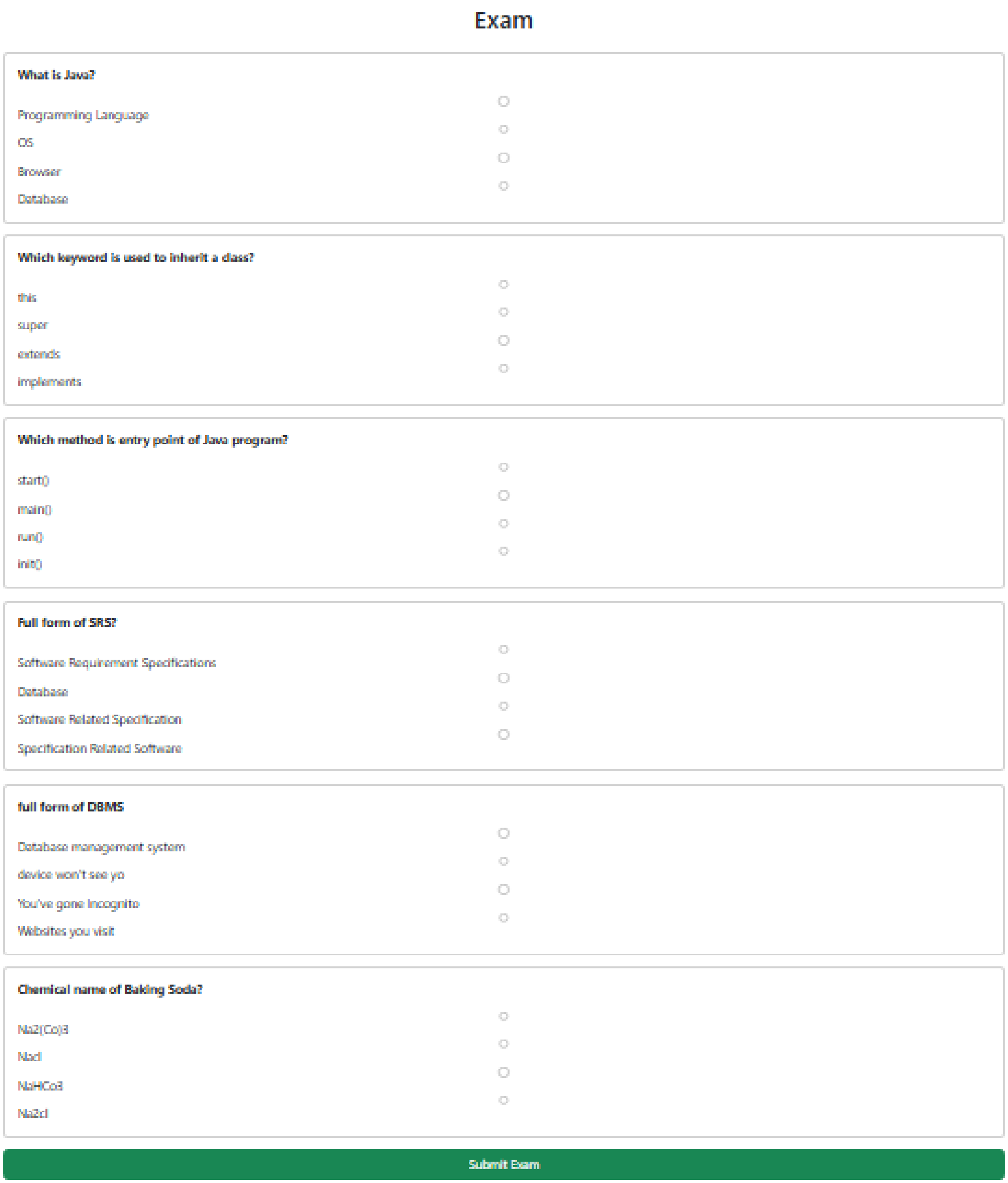
- Registration Page



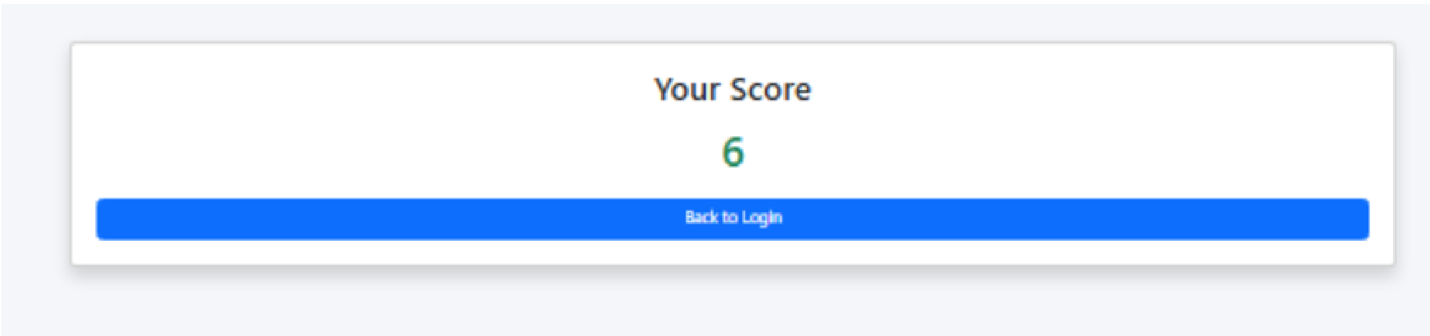
Student Registration Page of Online Examination System

- Allows new users to create an account
- User enters name, email, and password
- Input fields are validated
- Prevents invalid or empty entries
- Stores user data in database
- Provides secure access to system

- Exam Screen


Exam
What is Java?
Programming Language
OS
Browser
Database
Which keyword is used to inherit a class?
this
super
extends
implements
Which method is entry point of Java program?
start()
main()
run()
init()
Full form of SRS?
Software Requirement Specifications
Database
Software Related Specification
Specification Related Software
full form of DBMS
Database management system
device won't see yo
You've gone Incognito
Websites you visit
Chemical name of Baking Soda?
Na2(Co)3
Nacl
NaHCo3
Na2cl
Submit Exam

- Result Screen



Test Cases for Online Examination System

Test Case ID	Test Scenario	Input	Expected Output	Status
TC01	User Login	Valid username & password	Login Successful	Pass
TC02	User Login	Invalid credentials	Error message displayed	Pass
TC03	Student Registration	Valid details	Registration successful	Pass
TC04	Start Exam	Click "Start Exam"	Questions displayed	Pass
TC05	Answer Selection	Select correct option	Answer recorded	Pass
TC06	Submit Exam	Click "Submit"	Exam submitted successfully	Pass
TC07	Auto Submit	Time expires	Exam auto-submitted	Pass
TC08	Result Generation	After submission	Score displayed correctly	Pass
TC09	Admin Add Question	Valid question input	Question added successfully	Pass
TC10	View Result	Student clicks result	Result displayed	Pass

Maintenance Plan

The following maintenance activities will be performed to ensure the smooth functioning of the Online Examination System:

- **Bug Fixing:** Identify and fix errors or issues in the system regularly.
- **System Updates:** Update software features and improve functionality when required.
- **Database Maintenance:** Perform regular backup and recovery of data to prevent data loss.
- **Performance Monitoring:** Monitor system performance and optimize speed and efficiency.
- **Security Updates:** Implement security patches to protect user data and prevent unauthorized access.
- **User Support:** Provide assistance to users in case of technical issues.
- **Scalability Improvements:** Enhance the system to handle more users and data in the future.

References

The following resources were referred to during the design and implementation of the Online Examination System:

Books

1. Ian Sommerville, *Software Engineering*, 10th Edition, Pearson Education.
2. Herbert Schildt, *Java: The Complete Reference*, 11th Edition, McGraw Hill.
3. E. Balagurusamy, *Programming with Java*, McGraw Hill Education.

Websites

4. GeeksforGeeks – Java Programming & Concepts
   https://www.geeksforgeeks.org
5. Oracle Java Documentation – Official Java API Reference
   https://docs.oracle.com/javase
6. TutorialsPoint – Java Swing & JDBC Tutorials
   https://www.tutorialspoint.com
7. W3Schools – Basic Java & Database Concepts
   https://www.w3schools.com

Tools & Software

8. Visual Paradigm Community Edition – UML Diagram Tool
9. VS Code / IntelliJ IDEA – Development Environment
10. MySQL – Database Management System

Other References

11. Class Notes and Lecture Materials provided by faculty
12. Online tutorials and coding for Java Swing and JDBC implementation

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Citation Note

All the above resources were used only for learning and reference purposes. The implementation and design of this project are original and developed as part of academic coursework.

Conclusion

The **Online Examination System** developed in this project successfully demonstrates the design and implementation of a digital platform for conducting examinations in an efficient and secure manner.
The system automates key processes such as student registration, exam management, question handling, automatic evaluation, and result generation, thereby reducing the dependency on traditional manual methods.

Through this project, we were able to apply core concepts of **software engineering**, including requirement analysis, system design using UML diagrams (ER, Use Case, Sequence, and Class Diagrams), and implementation using Java and Swing.

The system provides a user-friendly interface for students to take exams and for administrators to manage question banks and results effectively.
The automatic evaluation feature ensures quick and accurate result generation, minimizing human errors and saving time.

Additionally, the system structure is designed to be scalable and maintainable, allowing future enhancements such as online deployment, advanced security mechanisms, and real-time monitoring.

Overall, this project highlights how modern software solutions can improve the examination process by making it more reliable, transparent, and accessible. It serves as a practical example of integrating theoretical knowledge with real-world application development.